

Linguistic Misinformation Markers

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Abstract

1 Introduction

We aim to build an annotated corpus that captures linguistic patterns distinguishing misinformation, opinion/editorial, from verified news. Our goal is to produce a dataset resource useful for researchers studying how language signals credibility — or the lack of it. We plan to enhance the Twitter Misinformation Dataset that is currently available on Hugging Face with the goal of contributing the enhanced dataset to Hugging Face.

Questions:

What linguistic markers distinguish misinformation from factual/verified information in news articles, news headlines and social media posts?

Is there a way to distinguish a third category of information?

Not all social media posts and news items are completely composed of fact or completely composed of misinformation. For example, a newspaper editorial likely contains facts, as well as opinions. Some social media posts are all about emotion without any facts, yet also without any misinformation.

Hypothesis: News and social media texts will have measurable differences in linguistic features — such as higher use of superlatives, emotional language, capitalization, vague sourcing, and clickbait phrasing, depending on whether the text aims to disseminate factual information, spread misinformation, or express emotions and opinions.

2 Methodology

2.1 Baseline Models

2.1.1 Traditional Method

For our traditional model, we used logistic regression. This model is appropriate for the text classification

task due to its scalability for sparse features and training speed.

To evaluate the model’s ability to identify opinions, text and numeric features were used. TF-IDF vectorization was applied over unigrams and bigrams for text data assessment. Additionally, our annotated linguistic features were scaled using StandardScaler before combining them with the TF-IDF matrix. The TF-IDF captures vocabulary and content patterns, while the annotated linguistic features identify writing style markers. Therefore, the inclusion of both feature types was necessary to adequately measure the differences between factual information, misinformation, and personal opinions.

For hyperparameter ‘C’, we used GridSearchCV to conduct 5-fold cross-validation on the training set. We used the macro F1 score to select the appropriate C value to account for class imbalances.

The results of the development set indicate that misinformation (Acc = 0.930; F1 = 0.905; AUC-ROC = 0.962) showed higher classification scores compared to opinions (Acc = 0.690; F1 = 0.683; AUC-ROC = 0.753).

2.1.2 Neural Method

For the neural model, a Convolutional Neural Network (CNN) was used in conjunction with NLTK’s tweetTokenizer and FastText frozen embeddings. Since the example texts in the corpus are of short length, opinion signals may exist in 3-gram, 4-gram, or 5-gram sequences, rather than in longer sequences. CNN was chosen because it relies on local markers by using a shifting window of n-grams to find patterns in text. In addition, given the low resource constraints of the project, CNN is robust against overfitting. Hyperparameters were chosen according to recommendations in “Convolutional Neural Networks for Sentence Classification,” (Kim, 2014) which recommends filter windows (h) of 3, 4, 5 with 100 feature maps each,

dropout rate (p) of 0.5, and l2 constraint (s) of 3.

Both the tokenizer and word embeddings were chosen because of their specialization in transforming internet text. NLTK's tweetTokenizer was used because it was developed to handle text boundaries that are unique to Twitter, including hashtags and emojis. Finally, FastText embeddings were selected because they are robust against typos and slang by using subword n-grams. The FastText embeddings were frozen to prevent additional overfitting of the training data, since the training set only included 600 examples.

When the CNN neural method we used was evaluated on the development data set, it scored accuracy = 0.7150, Macro F1 = 0.7141, F1(not opinion) = 0.7299, F1(opinion) = 0.6984 and AUC-ROC = 0.7747.

Classifying text as opinion or not opinion requires attention to detail and nuance. With text limited to 560 characters, the model has very little text to use for its classification. Some examples of false positives, items that were predicted as opinion, but are actually not opinion are:

[17] 'Birds in a blizzard. We put out extra sunflower seeds since their usual food sources just got,Ä¶ https://www.instagram.c' [25] 'I'm getting used to seeing a layer of ash on my car each morning - and I'm in AB. We've only seen a red sun this week. " [30] 'Many states along the Mid-Atlantic and the East Coast have shelters open in response to Hurricane Sandy. Search for an o' [55] "@jihettly @esd2000 good morning! Yes I'm in the middle of the blizzard. I have plenty of food so I'm happy! Lol. Stay wa"

Perhaps the model learned that words that are not specific, in terms of quantity, like "usual" in example 17, "getting used to" in example 25, "many" in example 30 or "plenty" in example 55 are indicators of opinion. In addition, example 55 has words like "happy" and "Lol" which are emotion words and may be used to express opinion.

Some examples of false negatives, items that were predicted as not opinion, but are actually opinion include:

[2] 'His argument is correct. Hurricane Rita killed over 100 people on the road, drowned them in their own cars. pic.twitter.' [113] 'Photo Falsely Claims Obama Served Food To Hurricane Harvey Victims http://www.westernjournalism.com/photo-falsely-claims' [133] 'Team Trump's efforts to spin Mulvaney's quid pro quo confession are failing. Mick Mulvaney and Mike Pompeo were tasked w'

[321] "Here's everything you need to know about the InsurrectionAct https://t.co/BbfljFN3RO"

In these examples, the model may be using words like "correct" in example 2, "failing" in example 133, and "know" in example 321 as indicators of an absence of opinion.

Since this CNN model is a baseline neural model, some level of misclassification is expected, given the difficult task.

3 Data

We used the [Hugging Face Twitter Misinformation Dataset \(Minassian\)](#) and annotated 1,000 examples labeled with the following categories:

1. **adjectives:** Adjectives with suffixes -ful, -less, -ment, -ness, -ing, or -ible.
2. **all_caps:** Capitalised words that are not acronyms.
3. **exclamation_marks:** Any number of exclamation marks.
4. **hedging:** Words found in the [hedging lexicon \(hedging lrec\)](#).
5. **unk:** Profanity.

We removed duplicate text entries and also considered texts to be duplicates when only the URL was modified, likely due to tokens added when choosing to share to Twitter/X from a third party website.

To ensure the hand-annotated dataset was manageable, we limited text length to 560 characters (twice that of a tweet); however, we believe that our models can scale to longer text.

They are currently labeled with 0 (factual) or 1 (misinformation), however, we aim to expand this to 3 classes, incorporating opinion. Given this, we added a column for opinion, with 1 as opinion as 0 as not opinion. Opinion was determined through first person point of view texts, not headlines that conveyed opinion, that showed a stance or preference towards a circumstance.

In terms of the dataset split and the small size of the dataset, we decided on a 60-20-20 train-validation-test split. This was conducted using stratification, to ensure similar proportions of each main annotator (>10%), proportion of misinformation vs. not that is similar to the larger dataset (65% factual, 35% misinformation) and distribution of

opinion to reflect the overall 1000 examples of 41% opinion and 59% not opinion. The dataset splits are added as a column and the script sets a random seed to 344 to ensure reproducibility.

4 Experimentation

5 Results

6 Conclusion

7 Limitations

8 Ethical Considerations

References

hedging lrec. Hedge words. https://github.com/hedging-lrec/resources/blob/master/hedge_words.txt.

Yoon Kim. 2014. [Convolutional neural networks for sentence classification](#). *Preprint*, arXiv:1408.5882.

Roupen Minassian. Twitter misinformation dataset. <https://huggingface.co/datasets/roupenminassian/twitter-misinformation>.